

Systematic Credit Mini-Pipeline PM Memo

Samuel Moss · May 14, 2026

1 What we built

A long-only US High Yield carry strategy targeting **risk-adjusted return per unit of spread risk**, not absolute alpha. We rank bonds by OAS / OASD (spread compensation per year of spread duration) and overlay a regime classifier that scales total deployment up or down based on credit conditions; rebalanced monthly. Each month:

- Rank the eligible HY universe by OAS / OASD (higher = better-paid per unit of spread duration risk).
- Hold the top 20% inverse-DTS-weighted, subject to a 5% issuer cap and 20% sector cap.
- Scale total deployment to 100% / 66% / 33% in risk-on / neutral / risk-off, set by a walk-forward composite-z-score classifier over VIX, the 10Y–2Y curve, and the Chicago Fed NFCI.

Backtested January 2010 through December 2018 against the equal-weighted excess return of the eligible HY universe. Sensitivity cutoffs of 10% and 30% reported alongside the primary 20% to verify the strategy isn’t a knife-edge at the chosen cutoff; same signal, same overlay, same constraints, only the universe cutoff varies.

Pipeline ships with 14 tests covering data ingestion, point-in-time merge leakage, signal computation, cap enforcement, regime thresholds, and backtest signal-to-return lag. Single command (`python -m src.main`) runs end-to-end in ~10 seconds; full architecture in README.

2 Why it makes sense

Signal. OAS / OASD is the natural risk-adjusted ratio for credit: basis points of spread compensation per year of spread duration. We’re not asking “which bonds will rally?” We’re asking “which bonds are paying me the most per unit of spread risk I take?” **Documented carry alpha in HY (Houweling & van Zundert 2017; Israel et al. 2018)**, with the asymmetry, carry strategies harvest steady gains but accumulate sharp left-tail losses.

Overlay. HY carry has negative skew; regime scaling truncates left-tail exposure. Implementation: 60-month rolling z-score with one-month shift (strictly past-looking); thresholds fit on 2000–2009 composites at the 33rd / 67th percentile; leakage controls verified by `test_regime_thresholds_derived_from_config`.

Rejected. OAS alone (loads on CCC + duration). OASD alone (loads on long-dated quality). OAS / DTS (degenerate — $DTS = OASD \times OAS$, so ratio reduces to $1/OASD$). ML signal (PM-defensibility). 5-state regime (over-fit).

3 What this is not

Not an alpha-maximization strategy. Absolute return is not the optimization target; we expect lower absolute returns than the equal-weighted HY universe and that’s by design. The trade is lower absolute return for materially better risk-adjusted return and drawdown control. Aggregate Information Ratio (-0.36) is negative for the same reason: IR penalizes the regime overlay’s deliberate deviations symmetrically. Sharpe, Sortino, and Calmar are the relevant metrics.

Not a deployment-ready strategy. Transaction costs are not modeled, reported Sharpe / Sortino / Calmar are gross. The IR benchmark is the equal-weighted excess return of the eligible universe, not an external HY index (BAMLH0A0HYM2 was FRED-restricted in April 2026 to a rolling 3-year window due to ICE licensing agreement). No live-execution layer, no intra-month risk controls beyond the documented caps, no per-Cusip exit-reason ledger.

4 Performance

Primary comparison is Top 20 vs benchmark on risk-adjusted metrics; Top 10 and Top 30 are sensitivity. **Top 20 chosen as primary** on best Sortino (2.47) and Calmar (1.37) at higher absolute return than Top 30; Top 10 disqualified by an 8.57% sector-concentration drawdown during the 2015–16 energy stress that the issuer + sector caps couldn’t fully absorb at the smaller bond count.

Metric	Top 10%	Top 20% (primary)	Top 30%	Benchmark
Annualized return (%)	2.84	2.67	2.59	4.81
Annualized vol (%)	3.01	1.46	1.37	7.03
Max drawdown (%)	-8.57	-1.94	-1.86	-18.44
Sharpe	0.94	1.83	1.89	0.68
Sortino	0.98	2.47	2.34	0.99
Calmar	0.33	1.37	1.39	0.26
Hit rate	0.83	0.82	0.81	0.64
Monthly turnover	0.55	0.49	0.46	—

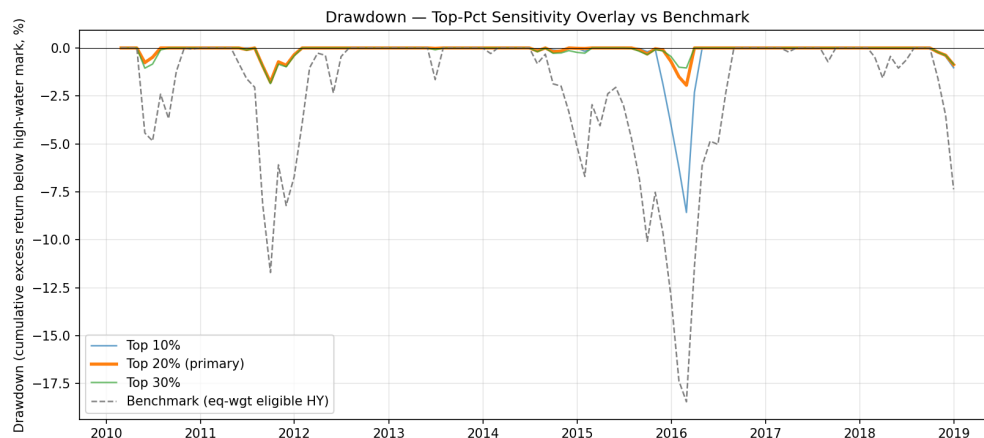
The bar chart in `outputs/charts/risk_adjusted_comparison.png` visualizes the same Sharpe / Sortino / Calmar dominance of the top-pct cutoffs that the table shows.

The risk-adjusted dominance holds at the regime level, not just in aggregate:

Regime	Top 20 Sharpe	Benchmark Sharpe	Top 20 Max DD (%)	Benchmark Max DD (%)
Risk-on	3.05	1.07	-0.76	-4.49
Neutral	2.10	0.58	-1.97	-11.78
Risk-off	0.85	0.63	-1.91	-11.58

Top 20 beats benchmark Sharpe in every regime, with max drawdown 4–6x smaller and the aggregate Sortino (2.47) and Calmar (1.37) advantages hold by construction at the regime level. Mean Top 20 exposure: Consumer Cyclical 17%, Communications 16%, Consumer Non-Cyclical 14%, Capital Goods 11%, Energy 9%. Rating: ~50% B / 26% BB / 24% CCC. Sector cap binds during Energy 2015–16 as designed.

5 Risk and Diagnostics



The regime overlay’s purpose is to truncate left-tail exposure during stress. The 2015–16 energy crisis is the test: benchmark drew down 18.44%, strategy drew down 1.94%. That’s the overlay doing its job mechanically: regime classifier flipped to risk-off, deployment scaled to 33%, drawdown contained at ~10% of benchmark’s. The Top 10 cutoff’s deeper 8.57% drawdown reflects sector concentration the caps couldn’t fully absorb at the smaller bond count.

Survivorship convention verified empirically (`test_exit_row_captures_terminal_event`): ~80% near-zero exit returns, ~1.5% catastrophic. Date convention validated at six anchor months against external HY index (inspection notebook A.8).

6 Failure modes & detection

- **Stale OAS / matrix pricing.** HY bonds trade infrequently; matrix-priced OAS lags real market state, anchoring the signal to stale prices. *Detection:* month-over-month $|\Delta OAS| < 1$ bp; baseline 1.75% in panel, alert above 5%.
- **Survivorship attribution.** Panel verified to record terminal-event returns in each exiting bond’s last row (`test_exit_row_captures_terminal_event`); `fill_value=0` at T+1 is not a material bias. *Detection:* per-Cusip exit-reason ledger reconciled against external HY default-rate data (Roadmap).
- **Regime classifier drift.** Macro indicators behave differently in production than during fit; regime labels become uninformative and position scaling unreliable. Already visible at mild level: backtest shows 47% neutral vs designed 33%, suggesting the 2000–2009 fit window (crisis-inclusive) is slightly over-conservative for the 2010–2018 expansion. *Detection:* rolling 12-month regime distribution; flag if any single label exceeds 80%, recalibrate threshold-fit window.

7 What we’d do next

- **Transaction cost modeling.** At ~49% monthly turnover with 50 bps round-trip HY bid-ask, gross 2.67% return faces ~2.9% annual cost drag (~1.9% scaled by mean deployment 0.66). Current cutoff selection is not viable net of realistic costs. Net-of-cost optimization may shift the preferred cutoff or rebalance frequency.
- **Signal vs. regime overlay ablation.** Carry-only run (allocation = 1.0) needed to decompose Sharpe / Calmar contribution between bond selection and regime scaling. Drawdown reduction is mechanically overlay-driven.
- **External benchmark cross-validation.** ICE BofA US HY Master II returned ~5.5–6% annually over 2010–2018 per Bloomberg; internal benchmark (4.81%) is order-of-magnitude consistent but composition-matched external comparison would strengthen attribution.
- **Per-Cusip exit-reason ledger.** Refines survivorship attribution beyond the current empirical `fill_value=0` convention.